ST FRANCIS CATHOLIC SECONDARY SCHOOL, LAGOS

The Jesuit-Run Catholic College

Basic Science and Technology 2

JSS 1

First Term

2026/2027

SCHEME OF WORK

WEEK	TOPICS	OBJECTIVES
1-2	UNDERSTANDING TECHNOLOGY	Concept of technology. Technology related careers
3	SAFETY GUIDELINES	Safety guidelines for pedestrians, cyclists and motorists. Road accidents, causes.
4	WORKSHOP SAFETY	Workshop accidents, causes types. Workshop safety rules and regulations. Safety devices. Accident prevention techniques in the workshop.
5-6	MATERIALS AND PROCESSING	Properties of materials: wood, metal, ceramics, plastics, rubber.
7-10	DRAWING PRACTICE	Lettering tools and application. Freehand sketching.
11-12	REVISION/EXAMINATION	

UNDERSTANDING TECHNOLOGY

CONCEPT OF TECHNOLOGY

Date	
Topic	**UNDERSTANDING TECHNOLOGY:** **CONCEPT OF TECHNOLOGY**
Learning Objectives	*i. define technology* *ii. state benefits of technology* *iii. mention and explain classes of technology* *iv. name products of technology* *v. apply technology to life issues*
Insight	*Understanding technology involves recognizing it as the application of scientific knowledge to solve problems, boost efficiency, and drive societal change. Key insights reveal how tech shapes our lives, from basic tools to AI, and why these concepts matter:*
Key words/New terms	*Technology, needs, products, developed, man.*
Application	The products we make use of from day to day such as chairs, books, tables, shelter, fans, etc, are products of technology.
Summary	Technology enables man to solve life problems. It makes work easier, faster and more accurate.
Student engagement: i. What have I learnt? ii. Skills acquired	
My Question(s)/Evaluation	i. ii. iii. iv. v.
Action	

Definition of technology

Technology is defined as the various methods and processes developed by man to enable him provide for his various needs and wants such as food, clothing, shelter, communication, transportation, etc.

Classes of Technology

There are two classes of technology. They are;

1. Underdeveloped technology
2. Highly developed technology.

Under Developed Technology

This involves the uses of manual process or the use of crude implements to do things. Manual process is the use of our hands to do things. Crude implements refer to hoes, axes, etc.

Highly Developed Technology

This involves the use of mechanized process or the use of machines and advanced tools to do work.

Benefits of Technology

The benefits of technology can be seen from the products and processes of production, such as:

1. Agriculture: modern farming equipment, fertilizers, storage facilities, etc.
2. Shelter: high rise buildings, skyscrapers, etc.
3. Security: armored car, guns, bombs, bullets, etc.
4. Healthcare: x-ray machine, digital weighing machine, stethoscope, etc.
5. Communication: radio, television, cell-phones, internet etc.
6. Transportation: motor vehicles, ships, air crafts, etc.
7. Education: magnetic marker board, computer, calculator, etc.

Therefore, the benefits of technology can be summarized as follows;

a. It provides solution to human problems, for example: technological research and development are used to find the cure for AIDS.
b. It creates new and easier ways of adding comfort to human lives.
c. It provides solutions to human being wants and desires. For example, desire to live in the other world planets are still in progress.
d. It provides means for skill acquisition and job creation.
e. It contributes positively to better all areas of human life.

Products of Technology

These refer to the materials, machines and equipment that are used to make our life easier. They are sometimes called Tools. They are so many and vary with specific functions. We can however, group some products as follows:

1. Products of technology found at Home: kitchen equipment, television, radio, electric fan, washing machine, etc.
2. Products of technology in Schools: computer, calculator, magnetic marker board, drum set, microphone, etc.
3. Products of technology found in the Office: air conditioner, laptop, fax machine, printer, photocopy machine, counting machine (money), internet facilities, CCTV, industrial generator, etc.

Technology and Society

Society: is a body of human beings living as members of a community.

The introduction of technology has brought a lot of improvement to our life and society. The society where products of technology are used is always modern in outlook. The standard of living of such a society is very high. The improvement brought on our society by technology includes the provision of social amenities such as:

1. Recreational parks and gardens.
2. Highway with traffic lights.
3. Modern building in urban landscape.
4. Treated drinkable water.
5. Modern transport facilities e.g. cars, trains, etc.

TECHNOLOGY RELATED CAREERS

The table below shows some engineering fields in technology and other related technological careers/fields in technology.

	Professions	Career	Job description
1.	Engineering	Software engineering	This is the writing of computer programs.
		Metallurgical and materials engineering	It is the technological study of metals, ceramics and their uses.
		Mechanical engineering	Design, produce, repair, service and maintain machines and devices.
		Aeronautical engineering	Study/work/researches related to space travel, aero plane, etc.
2.	Agriculture	Food science and technology	Handling of food using devices and computerized machines. E.g food preservation techniques.
3.	Technology	Information and communication technology	Information management, communication and wireless technology.
4.	Environmental	Architecture	Designing buildings either manually or with the use of Arch CAD (computer aided design).

		Estate management	Planning and managing estate location
		Meteorology	The science/study of the earth's atmosphere and its changes.

Role models in technology

Role models are good examples to others especially children that are still growing up. This group of young people admires the role model and wants to emulate them.

Examples of role model in technology in the locality are;

i. Mrs. Funmilayo Kuti – the first Nigerian woman to manipulate the technology of driving a car.
ii. Phillip Emeagwali – a Nigerian born computer scientist, one of two winners of the 1989 Gordon Bell prize from the Institute of Electrical and Electronics Engineering.

Date	
Topic	**SAFETY GUIDELINES**
Learning Objectives	*i. define safety guidelines* *ii. define pedestrian, cyclists and motorist* *iii. state safely guidelines for each of them* *iv. understand the causes of road accident* *v. mention examples of each group of road accident.*
Insight	*Effective road safety requires a shared responsibility between motorists, pedestrians and cyclists to protect the most vulnerable road users.*
Key words/New terms	*Guidelines, pedestrian, cyclist, motorist, crashes, accident.*
Application	Roads are found in our community and we make use of them to move from one point to another.
Summary	A pedestrian makes use of the legs; a cyclist makes use of the bicycle while a motorist makes use of the vehicle.
Student engagement: iii. What have I learnt? iv. Skills acquired	
My Question(s)/Evaluation	i. ii. iii. iv. v.
Action	

Safety guidelines are the processes of being safe.

Safety guidelines for Pedestrians

Pedestrian is the name given to someone walking in the street or along the road.

The following are some of the safety guidelines for pedestrians namely:

Use pavements or suitable footpaths provided.

i. It is best to avoid crossing at roundabouts where traffic approaches from all directions.
ii. Walk in the left side of the road, facing incoming traffic, where footpaths are provided.
iii. Make use of pedestrian crossing, traffic lights, subway, foot bridge or zebra crossing where provided to safely cross the road.
iv. Cross the road near lighted area at night so that you can be seen in the absence of pedestrian crossing.
v. Do not use your cell phone for calls and texting when crossing, use all your senses.
vi. Cross first to the island, then wait until the other part of the road is cleared to complete the crossing safely, that is where there is an island on the road.

Island: this is the central or middle portion of the road designed for refuge or safety in road crossing.

Safety guidelines for Cyclists

A cyclist is a person who rides a bicycle.

The following safety guidelines are applicable to all cyclists.

i. They must obey the same traffic rules and regulations as other vehicle drivers because they have the same right as other road users.
ii. Ensure that you do not cycle along the inside of large vehicles such as Lorries and buses, especially at junctions where most accidents take place.
iii. Ensure you wear a safety helmet.
iv. Ensure that your bicycle/motorcycle is in good condition particularly the brakes, lamps and rear reflectors before riding.
v. Do not carry more than one passenger on your motorcycle and never carry oversized objects.
vi. Obey the allowable maximum speed limit at all times and places.
vii. Keep safe distance from moving vehicle.
viii. Use lights and wear bright or reflective clothing, especially at night and during bad weather.

Safety guidelines for Motorists

A motorist is a person who drives a car.

The following are the safety rules and regulations for motorists. He/she must:

i. Obey all traffic regulations and road signals at all times.
ii. Perform the basic routine safety check on a motor vehicle daily before it is put into use.
iii. Fasten seat belt when driving to prevent you or your passenger form hitting the head on the dashboard in the event of a crash.
iv. Use side mirrors properly and regularly.
v. Obey the speed limit applicable on each road at all times.
vi. Ensure you drive to save lives, time and money.

Road accidents

Road accidents have claimed more lives and properties than major diseases. Road crashes can be grouped into these:

1. *Human factors:* these are found to contribute to about 90% of road accidents resulting from over speeding, lack of concentration, driving under the influence of alcohol and drugs, dangerous overtaking, etc.
2. *Mechanical factors*: this is due to irregular and poor maintenance of vehicles which leads to malfunctioning of the vehicles and loss of control and finally to road crashes. Examples of such mechanical factors are: malfunctioning of engine, brake failure, tyre burst faulty steering mechanism, electrical fault, etc.
3. *Environmental factors*: these are bad roads that result to road crashes. Examples are: potholes on the roads, slippery roads caused by flood, absence of road signals/road markings, Harmattan haze resulting to poor visibility, etc.

WORKSHOP SAFETY

Date	
Topic	**WORKSHOP SAFETY**
Learning Objectives	*i. define and identify a workshop* *ii. explain workshop safety and layout* *iii. be familiar with workshop safety rules and regulations* *iv. define workshop accidents* *v. state causes and precautions of workshop accidents.*
Insight	*Workshop safety is the proactive prevention of injuries, illnesses, and equipment damage through hazard identification, proper tool maintenance, and strict adherence to operating protocols. Your safety gear is your primary barrier against acute and chronic workshop injuries. Always assess the task and equip yourself with the appropriate safety protection.*
Key words/New terms	*Workshop, safety, regulations, devices*
Application	The workshop can be used for different purposes. Some examples around include the carpentry workshop, mechanic workshop, electrical workshop, etc.
Summary	The workshop is a room or building where technological products are made, repaired and maintained. Workshop safety is the condition of being free from any danger while working in the workshop.
Student engagement: v. What have I learnt? vi. Skills acquired	
My Question(s)/Evaluation	i. ii. iii. iv. v.
Action	

Definition of a Workshop

A workshop is a room/office specially built for caring out practical works such as construction, repair works or maintenance.

Workshop Safety

Workshop safety is the condition of being free of any danger or potential harm to the body while working in the workshop.

W**orkshop layout**

Workshop layout is the way in which machines, tools, work benches, etc are arranged in the workshop so that maximum safety can be achieved.

Rules and regulation in the workshop

1. Take and carry out instruction from the teacher
2. Do not touch any tool on display on the work bench unless asked to.
3. Do not play around in the workshop.
4. Do not operate any machine in the workshop unless asked to.
5. Return all used tools to their proper places after use.
6. Take precautions against accident by wearing safety devices and following laid down rules and regulations.
7. Make sure that the workshop is cleaned after use
8. Switch off all appliances before leaving the workshop.

Definition of Accident

An accident is an unpleasant occurrence that happens unexpectedly and causes pain, injury, damage or even death. Examples of workshop accidents are.

1. Stepping on a sharp metallic object
2. Having a cut injury
3. Splashing of corrosive chemicals on the body.
4. Ear problem due to sharp sound.
5. Fire injury or outbreak
6. Electrocution
7. Suffocation
8. Fall from height
9. Toxic gas emitting and inhalation.

Causes of Accident

The following may cause accidents in the workshop.

1. Lack of provision of safety devices
2. Failure to use safety devices where provided.
3. Failure to follow laid down methods for handling facilities.

Safety Devices

1. Protective clothing for the protection of the skin e.g. overalls, lab coats, safety jackets.
2. Eye goggles / eye shield / Eye visor for eyes protection.
3. Workshop shoes/boots safety boots for feet protection
4. Hand gloves for hand protection.
5. Nose cover for nose protection.
6. Helmet for head protection in case of fall or object hitting the head, etc.

Safe Working Methods/Accident prevention techniques

The following are how to prevent accidents while working in the workshop;

1. Always use the right tool for each job.
2. Always keep to instruction.
3. Do not work on oily, greasy and wet floor in order to prevent injury/slipping.
4. Always keep away chipped and sharp objects in order to prevent injury.
5. Do not use worn out tools.
6. Do not welcome distraction while working.
7. Always use your imagination while doing sensitive work.

Types of Fire

1. **Electrical fire:** this arises from electrical sources e.g. electric sparks, partial contact when items are plugged in sockets or burning of appliances.
2. **Chemical Fire:** any fire where combustible chemicals, flammable liquids, or reactive substances are the primary fuel source such as spirit, petrol, diesel and solid fuels such as wood, paper, cloth etc.

Fire-fighting equipment

1. **Fire Extinguisher:** These are cylindrical metals painted red and containing chemicals for putting out small fires.
2. **Sand Buckets:** these are metal buckets pointed red and piled with dry sand for putting out small fires.
3. Fire blankets.
4. **Fire Alarm:** This is a siren to alert people when there is fire out-break.

MATERIALS AND PROCESSING

Date	
Topic	**MATERIALS AND PROCESSING**
Learning Objectives	*i. identify and classify wood* *ii. identify and classify metals* *iii. identify ceramics and glass* *iv. identify and classify plastics* *v. identify and classify rubber*
Insight	*Identifying materials (wood, metal, ceramics, plastic, rubber, and glass) relies on observing key physical and chemical properties such as density, thermal conductivity, elasticity, and reaction to heat. Each material exhibits unique behaviors that make it distinct and suited for specific applications*
Key words/New terms	*Wood, metal, ceramics, glass, rubber, plastic, properties.*
Application	The products we make use of are made of different materials such as wood, metals, ceramics, plastics and rubber.
Summary	The different materials used for making technological products can be identified using their properties. They can also be classified based on these properties.
Student engagement: vii. What have I learnt? viii. Skills acquired	
My Question(s)/Evaluation	i. ii. iii. iv. v.
Action	

Properties of Materials– WOOD

Introduction

An engineering material is a substance from which objects such as tools, machines, furniture, home appliances and other equipment are made.

Identification of Wood

Wood is the solid portion of a tree stem found below the outer covering of the stem. After it is prepared for use, it is referred to as timber. It is not every tree that can supply timber. Only those trees whose stems get wider in cross-section, roughly on a yearly basis can produce timber. Iroko, Opepe, etc belong to this group. Some others whose stem increasing in cross-section after some period of growth do not supply timber. Bamboo, canes, etc, belong to this group.

Classification of wood

There are two classes of wood or tree. They are;

1. Hard wood (deciduous trees)
2. Soft wood (coniferous trees)

They are distinguished based on their properties in the table below.

	Hardwood	**Soft wood**
1	They are deciduous woods	They are coniferous wood
2	They are trees with broad leaves	Their trees have needlelike leaves which do not shed their leaves and so called **evergreens.**
3	Timber is called hard-wood timber	Timber is called soft-wood timber
4	Usually found in the temperature and tropical zones of the world	Usually found in temperate or colder zones of the world.
5	Trees have short trunks and bushy heads.	Trees have cone –shaped heads and straight tape-ring stems.
6	Seeds are contained in seed cases.	Seeds are contained in cones.
7.	It is dark-colour in nature	They are light in colour.
8	They have complex cellular structure Examples Iroko, mahogany, Obeche, Agba,Oak, Teak, Afara, Cherry, Walnut trees, etc.	They have simple cellular structure. Examples Spruce, pine, cedar, cypress, fir, etc.

Properties of Wood

1. Colour: wood appears in dark brown or light colour.
2. Hardness: timber is classified into three namely: soft, moderately hard, hard.
3. Specific gravity or density: timber can be classified by the measurement of their density into light, moderately heavy and heavy.

4. Electricity conductivity: wood is a poor conductor of electricity when dry but good conductors when wet.
5. Polish effect: wood appears shining when polished.
6. Moisture content: timbers have different moisture content at different times of seasoning. Variations in moisture content affect the properties and strength of the timber.

Properties of materials – METAL

Classes of Metal

There are two classes of metal. They are;

1. Ferrous Metals: they are those metals that contain high percentage of iron and are very magnetic. Examples are cast iron, wrought iron and steel.
2. Non-ferrous metals: they are those metals that do not contain iron and therefore not magnetic. Examples are aluminum, zinc, copper, lead, tin, etc.

Alloys (artificial metals)

An alloy is a mixture of two or more metals. The resulting alloy can have different properties and qualities from the original metals. This is very useful because alloys can be made with special properties for specific purposes.

Classes of alloys

1. Ferrous Alloys: these are alloys produced from ferrous metals e.g stainless steel, nickel steel, etc.
2. Non-ferrous Alloys: These are alloys produced from non-ferrous metals e.g bronze, brass, etc.

Identification of metals

Metals can be identified through the following properties.

1. Density: This is the ratio of mass of the substance per unit volume.
2. Colour/Lustre: When metals are polished, they begin to shine or glitter.
3. Fusibility: Metals always melts when they are heated.
4. Magnetic property: Metals have the ability to be attracted to magnet, either strongly or weakly, when they are in contact with one.
5. Ductility: This is the ability of metals to be drawn into thin wires.
6. Toughness: This is the ability of a metal to resist stress, breakage or cracking.
7. Conductivity: This is the ability of a metal to transfer heat and electricity.
8. Sound test: All metals produce sounds when they are struck.
9. Malleability: this is the ability of a metal to be beaten into shapes.
10. Corrosive: this is the ability of a metal to corrode or rust easily.

Forms of Metals

Metals can be found commercially in different forms such as rods, section, pipes, wires, sheets, bars, plates and channel sections.

PLASTIC

Identification properties

1. Plastics are non-metallic and organic materials.
2. Plastics are derived from petroleum products.
3. At high temperatures, plastics flow like liquid and are then easily moulded and shaped.
4. Plastics are generally light in weight.
5. Plastics do not rust or sound.
6. Plastics have good resistance to oil, grease and most common solvents.
7. Plastics are poor conductors of heat and electricity.
8. Plastics possess low strength.

Classes of plastics

There are two types of plastics. They are:

1. Thermoplastics
2. Thermosets.

1. **Thermoplastics:** this plastic cannot withstand heat. When heated, it becomes molten and can be re moulded again. E.g. plastic buckets, jerry cans, cup, etc.
2. **Thermosets:** This plastic can withstand heat. When heated, it burns and cannot be re moulded again e.g. shirt button, pot handles, knife handled, etc.

Rubber

There are two types of rubber. They are;

1. Natural rubber: This is produced from latex of rubber trees. Natural rubber reacts to sunlight. It is therefore improved by a process called vulcanization. It involves adding sulphur and carbon black to the rubber and heating it, which makes it harder and more heat resistant.
2. Synthetic rubber: It is obtained from organic materials derived from petroleum. They have better qualities than natural rubbers. They are resistant to sunlight, temperature changes, chemicals and do not age like natural rubber.

Properties of rubber

They are:

i. Elastic.
ii. Poor conductors of electricity.
iii. Shock-absorber and used as engine mountings.
iv. Waterproof.

Ceramics and glass

Ceramics and glass are objects made from earthen materials and hardened by heat. E.g. clay pots, tiles, blocks, glass cup, glass bottle, etc.

Properties

Ceramics and glass are:

1. Brittle: ceramics and glass easily break.
2. Heat resistant: ceramics and glass have high melting point.
3. Inorganic material: ceramics and glass are produced using artificial chemicals.
4. Non-metallic: ceramics and glass are insulators. i.e. poor conductor of heat and electricity
5. Hard in shape and appearance.

Differences between Ceramics and Glass

	Ceramics	**Glass**
1.	It is not transparent.	It is transparent
2.	It is dense.	It is less dense.
3.	it cannot be processed into mirror.	It can be processed into mirror.

DRAWING PRACTICE

Date	
Topic	**DRAWING PRACTICE**
Learning Objectives	*i. define drawing* *ii. identify and make use of drawing instruments and materials* *iii. care for the drawing instruments and materials* *iv. define free hand sketching* *v. draw simple shapes using free hand*
Insight	*Drawing instruments and materials provide the physical foundation for precise technical drafting, geometric design, and creative art. They are broadly categorized into **reusable instruments** (tools that assist with precision and accuracy) and **consumable materials** (the mediums and surfaces being marked upon).* *Freehand sketching is the rapid non instrumental translation of ideas into 2D OR 3D visual representations. It is the essential first step in design—fostering cognitive problem-solving, enhancing spatial reasoning, and enabling designers to iterate ideas before committing to precise, time-consuming drafts.*
Key words/New terms	*Drawing, instruments, materials, freehand.*
Application	Some common drawing instruments and materials are drawing board, compass, ruler, divider, drawing paper, pencil, etc. Drawing is applicable in different areas such as engineering, agriculture, tailoring, etc.
Summary	Drawing is a means of communication. It is done using various instruments and materials. Freehand sketching is a type of drawing where no any other instruments are used except pencil and an eraser.
Student engagement: ix. What have I learnt? x. Skills acquired	

My Question(s)/Evaluation	i. ii. iii. iv. v.
Action	

Introduction to Technical Drawing

Drawing as a means of communication affords common understanding universally, while writing is done in various languages and it is only one who understands it that can interpret it. Drawing is important as it is often easier to explain details by drawing or sketches them by written description. Drawing thus forms an effective means of communication.

Technical drawing is a means of communication. It provides a system or orderly rules and regulation which enable a man called a designer to draw his ideas on paper.

Identification of Drawing Instrument/Material.

1. Drawing board: it is rectangular in shape and of different sizes. The normal size of a drawing board is A2 (650mm by 470mm). the surface of the drawing board must be smooth because the drawing paper is placed on it.

2. T-square: it is made of metal, plastic or hardwood with or without transparent edges. The head of the tee square is normally fixed and the working edge of the blade must be straight. T-square is used for drawing horizontal, vertical, inclined or parallel lines on the drawing paper.

3. Protractor: it is used for drawing or measuring any desired angle. They are made of hard transparent plastic or celluloid and are usually semi-circular in shape.

4. Compass: a compass is used for drawing circles and arcs. it consists of two legs hinged together at its upper end, a pointed needle fitted at the lower end of one leg and a pencil which is inserted at the end of the second leg. The pencil is removable.

5. Set-squares: these are transparent triangular shaped instruments used with a T-square or a metric rule for drawing vertical or inclined lines. There are two types of setsquares:

- **60°× 30° setsquare**
- **45° setsquare**

6. French curve: it is used to draw irregular curved lines which cannot be drawn with a compass. It is made of transparent celluloid or plastic and available in various shapes and sizes.

7. Divider: a divider is similar to a compass, except there is no provision for pencils. It is used for setting off distances and transferring measurement.

8. Sharpener

9. Templates

10. Scale rule

Identification of Drawing Materials

1. Pencils: this is the most essential drawing material. Good quality pencils must be well sharpened to a conical point shape for good quality drawings. The grades of pencils available are**: H grade pencils which are hard and are used for light lines.**

B grade pencils are soft and darker, used for art work.

HB pencils are used for sketching, lettering and dimensioning.

2. Drawing papers: this is where the drawing is carried out. Good quality **A4** paper with smooth surface is recommended for use in class work and must be attached to the drawing board with adhesive tape.
3. Eraser
4. Adhesive tape
5. Sand paper: used for smoothening pencil tips.
6. Dusting brush: used for removing eraser particles.

The Distinction Between Tools & Materials

- **Drawing Instruments:** Reusable, non-consumable tools used to guidelines, measure distances, and construct geometric shapes. They do not get used up in the drawing process.
- **Drawing Materials:** Consumable items and surfaces that are depleted or used up during the drafting and colouring/shading process.

Care of Instruments

1. Do not use pin on the drawing board.
2. Do not use the drawing board for any other purpose except drawing
3. When not in use, place the drawing board face down.
4. Do not use blade or any sharp object on the tee-square or set-square
5. All instruments must be kept clean
6. Dividers or compass points must never be used as paper holders.
7. Drawing instruments should be kept in their packets after use.

LETTERING TOOLS AND APPLICATION

The lettering used on drawing is the single stroke Gothic letters and numerals as shown in your textbook. Lettering on drawings must be legible and easy to understand. It is recommended that only one style should be used throughout a drawing. The pencil good for lettering is the HB pencil sharpened to a conical shaped tip.

ACTIVITY (when constructing the title block and other given examples in the textbook)

BOARD PRACTICE

It is the term used for drawing layout on the paper. This means the steps to follow in order to get your drawing sheet properly positioned on the drawing board.

Step 1: Place your half imperial drawing board conveniently on a flat table.

Step 2: Place your drawing sheet close to the upper position.
Step 3: Place your tee square horizontally at the lower part of the drawing board with the stock at the left hand side edge and align the lower edge of the drawing sheet with tee-square.
Step 4: Fix firmly the drawing sheet to the drawing board with masking tapes, each at the four corner of the drawing board.
Step 5: Draw the border line 10mm round the four edges of the drawing board using tee-square to draw the horizontal lines and using set-square to draw the vertical lines.

Drawing Border lines

Border lines are also called Margins. A space of 10mm is left all around the drawing paper and is demarcated with straight lines. These lines are called Border lines. They beautify the drawing paper and protect the drawing.

Title block

After the drawing has been laid on paper, it is necessary to give some information to enable others understand what the drawing is about. Such information relating to the name of the designer, school, class, date, title, scale and date is put in what is known as TITLE BLOCK. The title block is usually at the bottom right-hand corner of the drawing paper.

ACTIVITY- PRACTICE IN CLASS

FREE HAND SKETCHING

Free hand sketching is a quick and convenient way of describing the shape of an object. In free hand sketching no instrument is used except a pencil or pen and probably an eraser.
Free hand sketching is a quick way of passing across information.

TECHNIQUES FOR SKETCHING

To draw a straight line

Plot some points not far from each other. Now, join the points together to form a straight line.

To plot a curve

Plot some points not far from each other but at different levels. Joint the points to form a curve.

Circle

Draw lines which are equal in diameter to the circle in different directions. Each line must be drawn as faint and straight as possible, each crossing one another at a central point. Now, join the points by little curves from the tip of each line.

Irregular edges

Figures with irregular edges should first be fixed in a rectangular or square shape which can completely contain the object. The shape is then carefully shaped by using dotted lines joined to show the desired shape with the appropriate curves.

ACTIVITY- PRACTICE IN CLASS